Notes: Bhamra and Uppal (AER, 2019)

Does Household Finance Matter? Small Financial Errors with Large Social Costs

Contents

1 Big picture

- **Question.** How costly is household portfolio underdiversification once we move beyond a static mean-variance calculation and allow it to affect asset allocation, intertemporal consumption, and aggregate growth?
- **Setting.** The paper builds a dynamic production-economy model with many firms and many households. Households have Epstein–Zin preferences and familiarity biases, so they tilt their portfolios toward a few "familiar" risky assets.
- **Core challenge.** Existing household-finance evidence shows that diversification mistakes look modest in a static portfolio problem. The paper asks whether this conclusion survives once those mistakes feed back into saving, the risky share, and macroeconomic growth.
- **Main result.** The direct mean-variance loss from underdiversification is small, roughly comparable to giving up about 1 percent per year in portfolio return. But once familiarity biases distort the risky share and the consumption-savings decision, household welfare losses become about four times larger.
- **General-equilibrium result.** Even when familiarity biases cancel out in aggregate portfolios, they do *not* cancel out in aggregate saving decisions. The induced change in saving distorts capital accumulation and growth, pushing social welfare losses above the private mean-variance losses by more than a factor of five.
- **Main intuition.** Familiarity bias does not just create a "bad portfolio." It also makes households hold too little of the risky production technology and choose the wrong consumption path over time. Aggregating those mistakes changes the growth rate of the whole economy.
- **Why this paper matters.** It is a clean example of how seemingly small household financial mistakes can have first-order social costs. The paper is therefore an argument that household finance is not a sideshow to macroeconomics.

2 Data and measurement (key objects)

This is a **theoretical and calibrated** paper rather than a reduced-form empirical paper. So the relevant "data" are the model objects and the empirical calibration targets used to discipline magnitudes.

2.1 Production economy and risky assets

- There are N risky firms and H households.
- Firm n has capital stock $K_{n,t}$ and pays out investment/dividend flow $D_{n,t}$.
- Capital evolves according to

$$dK_{n,t} = (\alpha K_{n,t} - D_{n,t})dt + \sigma K_{n,t}\, dZ_{n,t}.$$

- **Interpretation.** All firms have the same expected return α and the same idiosyncratic volatility σ, but shocks are firm specific. Therefore diversification changes risk, not mean return.

2.2 Household wealth and portfolio objects

- Household h allocates wealth $W_{h,t}$ between a risk-free asset $B_{h,t}$ and the N risky firms.
- Wealth decomposition:

$$W_{h,t} = B_{h,t} + \sum_{n=1}^{N} K_{hn,t}.$$

- Let $\omega_{hn,t}$ be the share of wealth invested in risky firm n.
- The share of total wealth invested in all risky assets is

$$\pi_{h,t} = \sum_{n=1}^{N} \omega_{hn,t}.$$

- The within-risky-portfolio weight on firm n is

$$x_{hn,t} = \frac{\omega_{hn,t}}{\pi_{h,t}}.$$

- The consumption-wealth ratio is

$$c_{h,t} = \frac{C_{h,t}}{W_{h,t}}.$$

- **Interpretation.** The model separates three choices: consumption-saving ($c_{h,t}$), risky-versus-safe allocation ($\pi_{h,t}$), and diversification across risky firms ($x_{hn,t}$).

2.3 Preferences and familiarity-bias objects

- Households have Epstein–Zin preferences with:
  - subjective discount rate δ_h,
  - elasticity of intertemporal substitution ψ_h,
  - relative risk aversion γ_h.
- Familiarity with firm n is $f_{hn} \in [0, 1]$.
- Larger f_{hn} means the household feels more familiar with the asset.
- The average familiarity of household h is

$$\mu_{fh} = \frac{1}{N} \sum_{n=1}^{N} f_{hn},$$

  and the cross-firm variance of familiarity is

$$\sigma_{fh}^2 = \frac{1}{N} \sum_{n=1}^{N} (f_{hn} - \mu_{fh})^2.$$

- **Interpretation.** μ_{fh} matters for the household's overall willingness to hold risky assets, while σ_{fh}^2 matters for how tilted the household is inside the risky portfolio.

2.4 Calibration targets

- The key discipline comes from Calvet, Campbell, and Sodini's evidence on Swedish households.
- Volatility of the median underdiversified household portfolio:

$$\sigma_x = 19.5\% \text{ per year.}$$

- Volatility of the same household under full diversification:

$$\sigma_{1/N} = 14.7\% \text{ per year.}$$

- Equity premium:

$$\alpha - i = 6.7\% \text{ per year.}$$

- Risk-free rate used in calibration:

$$i = 3.7\% \text{ per year,} \qquad \alpha = 10.4\% \text{ per year.}$$

- Baseline preference parameters:

$$\delta = 0.03, \qquad \psi = 1.20.$$

- In partial equilibrium, matching the typical risky share $\pi_h = 0.53$ implies

$$\gamma_h = 3.3245.$$

- In general equilibrium, imposing market clearing with $\pi_h = 1$ implies a lower economy-wide risk aversion:

$$\gamma = 1.762.$$

- **Interpretation.** The paper is not trying to estimate the whole model from scratch. It is asking: *given empirically realistic underdiversification, how large are the implied welfare losses once dynamic and macro channels are turned on?*

2.5 Unobservables and timing

- There is no firm panel or household panel to estimate reduced-form coefficients.
- The key unobservable is the *degree of familiarity* with each firm, represented by f_{hn}.
- Time is continuous, which helps deliver closed-form certainty-equivalent and portfolio expressions.
- The model distinguishes:
  - **partial equilibrium:** individual household decisions holding macro growth fixed;
  - **general equilibrium:** aggregate saving determines growth and therefore feeds back into welfare.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production side and household budget set

Each risky firm has a capital stock evolving as

$$dK_{n,t} = (\alpha K_{n,t} - D_{n,t})dt + \sigma K_{n,t}\, dZ_{n,t}.$$

Household wealth is split between the bond and all risky firms:

$$W_{h,t} = B_{h,t} + \sum_{n=1}^{N} K_{hn,t}.$$

The risk-free asset earns

$$\frac{dB_{h,t}}{B_{h,t}} = i\, dt.$$

The dynamic budget constraint is

$$\frac{dW_{h,t}}{W_{h,t}} = (1 - \pi_{h,t})i\, dt + \pi_{h,t} \sum_{n=1}^{N} x_{hn,t}(\alpha\, dt + \sigma\, dZ_{n,t}) - c_{h,t}\, dt.$$

- **Interpretation.** Wealth growth has three sources: the safe return, the risky portfolio return, and consumption drag.
- Because all risky firms have the same expected return, the only reason to diversify is to reduce idiosyncratic risk.

3.2 Preferences and certainty equivalent

The Epstein–Zin time aggregator is

$$\mathcal{A}(a, b) = \left[(1 - e^{-\delta_h dt})a^{1-\frac{1}{\psi_h}} + e^{-\delta_h dt} b^{1-\frac{1}{\psi_h}}\right]^{\frac{1}{1-\frac{1}{\psi_h}}}.$$

The certainty equivalent of utility one instant ahead is

$$\mu_{h,t}[U_{h,t+dt}] = \mathbb{E}_t[U_{h,t+dt}] - \frac{1}{2}\gamma_h U_{h,t} \mathbb{E}_t\left[\left(\frac{dU_{h,t}}{U_{h,t}}\right)^2\right].$$

- **Interpretation.** This is the usual "expected utility minus a risk penalty" representation in continuous time.
- The risk penalty scales with relative risk aversion γ_h and the variance of proportional utility changes.

3.3 Modeling familiarity bias as ambiguity

The familiarity-biased certainty equivalent is

$$\mu^{\nu}_{h,t}[U_{h,t+dt}] = \mu_{h,t}[U_{h,t+dt}] + U_{h,t}\left(\frac{W_{h,t}U_{Wh,t}}{U_{h,t}}\,\pi_{h,t}\nu_{h,t}^{\top}x_{h,t} + \frac{1}{2\gamma_h}\frac{\nu_{h,t}^{\top}\Gamma_h^{-1}\nu_{h,t}}{\sigma^2}\right)dt,$$

where $\nu_{h,t}$ is the vector of return adjustments chosen by the household and Γ_h is diagonal with entries $(1-f_{hn})/f_{hn}$.

- **Interpretation.** A household acts as if it marks down the expected returns of unfamiliar assets.
- The first new term changes perceived expected return.
- The second new term penalizes large adjustments, especially for familiar firms.
- Hence familiarity is modeled as *less ambiguity* about some assets than others.

3.4 The household problem

Without familiarity bias, the household solves

$$\sup_{C_{h,t}} \mathcal{A}\left(C_{h,t}, \sup_{\pi_{h,t},x_{h,t}} \mu_{h,t}[U_{h,t+dt}]\right).$$

With familiarity bias, it solves

$$\sup_{C_{h,t}} \mathcal{A}\left(C_{h,t}, \sup_{\pi_{h,t},x_{h,t}} \inf_{\nu_{h,t}} \mu^{\nu}_{h,t}[U_{h,t+dt}]\right).$$

Under homotheticity and constant returns to scale, the HJB reduces to

$$0 = \sup_{C_h}\left(\delta_h u_\psi\left(\frac{C_h}{U_h}\right) - \frac{C_h}{W_h} + \sup_{\pi_h,x_h}\inf_{\nu_h} MV_h(\pi_h,x_h,\nu_h)\right),$$

with

$$MV_h(\pi_h,x_h,\nu_h) = [i+(\alpha-i)\pi_h] - \frac{1}{2}\gamma_h\sigma^2\pi_h^2 x_h^{\top}x_h + \pi_h\nu_h^{\top}x_h + \frac{1}{2\gamma_h}\frac{\nu_h^{\top}\Gamma_h^{-1}\nu_h}{\sigma^2}.$$

- **Interpretation.** The problem cleanly decomposes into:
  - an intertemporal consumption problem;
  - a static mean-variance problem with familiarity-adjusted expected returns.
- This decomposition is the key analytical trick of the paper.

3.5 Optimal diversification inside the risky portfolio

The optimal return adjustment is

$$\nu_{hn} = -(\alpha - i)(1 - f_{hn}).$$

So the more familiar the asset, the smaller the pessimistic adjustment.

The optimal within-risky-portfolio weights are

$$x_{hn} = \frac{f_{hn}}{N\mu_{fh}}.$$

The risky-portfolio variance is

$$\sigma_{x_h}^2 = \sigma_{1/N}^2 \left[1 + \left(\frac{\sigma_{fh}}{\mu_{fh}}\right)^2\right].$$

- **Interpretation.** Familiarity bias mechanically pushes the household away from the equal-weighted risky portfolio.
- If all $f_{hn} = 1$, then $x_{hn} = 1/N$ and the household holds the diversified benchmark.
- The larger is the cross-firm variance of familiarity, the larger is portfolio risk.

3.6 Optimal risky share

The share of wealth invested in risky assets is

$$\pi_h = \frac{\alpha - i}{\gamma_h \sigma_{1/N}^2} \mu_{fh}.$$

Without familiarity bias,

$$\pi_h^U = \frac{\alpha - i}{\gamma_h \sigma_{1/N}^2}.$$

- **Interpretation.** Familiarity bias does not only alter *which* risky firms the household holds.
- It also lowers *how much* wealth the household puts in risky assets at all.
- This is a central message of the paper: underdiversification creates excessive risk, and the household partially responds by under-investing in the productive risky sector.

3.7 Mean-variance welfare

The mean-variance welfare of the household is

$$U_h^{MV}(\pi_h, x_h) = [i + (\alpha - i)\pi_h] - \frac{1}{2}\gamma_h \pi_h^2 \sigma_{x_h}^2.$$

- **Interpretation.** Familiarity bias hurts welfare through a worse risk-return trade-off.
- The paper emphasizes two pieces:
  - higher variance from bad diversification;
  - lower expected return because the household scales down the risky share.

3.8 Consumption rule and lifetime utility

The optimal consumption-wealth ratio is

$$c_h = \psi_h \delta_h + (1 - \psi_h) MV_h(\pi_h, x_h, \nu_h).$$

- **Interpretation.** Because familiarity changes the perceived return-risk trade-off, it also changes the household's optimal saving rate.
- If $\psi_h > 1$, greater risk tends to raise current consumption and reduce saving.
- If $\psi_h < 1$, the income effect can dominate and saving can move the other way.

Lifetime utility per unit of wealth can be summarized by

$$\frac{U_{h,t}}{W_{h,t}} = \kappa_h,$$

where κ_h is a nonlinear function of δ_h, ψ_h, U_h^{MV}, and c_h.

- **Interpretation.** Once consumption is endogenous, familiarity bias matters not only through the static portfolio problem but through the whole intertemporal path of consumption.

3.9 Aggregate restrictions used to isolate the macro channel

The paper imposes two restrictions.

First, aggregate risky-portfolio tilts cancel out:

$$\frac{1}{H} \sum_{h=1}^{H} \epsilon_{hn} = 0 \qquad \forall n,$$

where ϵ_{hn} is the deviation of household h's risky-portfolio weight on firm n from the unbiased benchmark $1/N$.

Second, all households have the same average familiarity:

$$\mu_{fh} = \mu_{fj} = \mu_f \qquad \text{for all } h, j.$$

- **Interpretation.** These restrictions guarantee that the paper's general-equilibrium effects are *not* driven by some firms being overfunded in aggregate.
- Instead, the macro effect comes from distorted *saving* and *growth*, even when aggregate portfolio tilts cancel out.

3.10 Market clearing, the risk-free rate, and growth

Bond market clearing with zero net supply implies the equilibrium risk-free rate

$$i = \alpha - \gamma \frac{\sigma_{1/N}^2}{\mu_f}.$$

The common consumption-wealth ratio in equilibrium is

$$c = \psi\delta + (1-\psi)\left(\alpha - \frac{1}{2}\gamma\sigma_x^2\right).$$

Aggregate growth is

$$g = \frac{I_t^{agg}}{K_t^{agg}} = \alpha - c = \psi(\alpha-\delta) - \frac{1}{2}(\psi-1)\gamma\sigma_x^2.$$

- **Interpretation.** Better diversification lowers σ_x^2.
- If $\psi > 1$, that raises saving and increases growth.
- If $\psi < 1$, growth can move the other way, but welfare still rises because the reduction in risk dominates.

3.11 Social welfare and the micro-versus-macro decomposition

Aggregate social welfare can be written as

$$U_t^{social} = \kappa K_t^{agg},$$

where κ depends on preferences and on the mean-variance term

$$U^{MV}(\sigma_x) = \alpha - \frac{\gamma}{2}\sigma_x^2.$$

The total social-welfare effect of reducing portfolio risk can be decomposed as

$$\frac{dU^{MV}(\sigma_x)}{d(\sigma_x^2)} = \frac{\partial U^{MV}(\sigma_x)}{\partial(\sigma_x^2)} + \frac{\partial U^{MV}(\sigma_x)}{\partial g}\frac{\partial g}{\partial(\sigma_x^2)}.$$

The paper shows this yields

$$\frac{d\ln\left(U_t^{social}/K_t^{agg}\right)}{d(\sigma_x^2)} = -\frac{1}{2}\frac{\gamma}{c}\left(\frac{1}{\psi} + 1 - \frac{1}{\psi}\right).$$

- **Interpretation.** The first term is the **micro** effect: lower portfolio risk directly raises utility.
- The second term is the **macro** effect: lower risk changes saving and growth.
- When $\psi = 1$, the macro term vanishes exactly.
- When $\psi > 1$, the macro amplification is positive and can be large.

3.12 Robustness and discussion: labor income

The discussion section asks whether background risk overturns the main results.

- With deterministic or stock-return-uncorrelated labor income, households optimally want to hold *more* risky assets, which would strengthen the relevance of bad diversification.

- If labor income is positively correlated with employer stock returns, the case for diversification becomes even stronger.
- Only if labor income were negatively correlated with employer stock returns would familiarity bias become less costly, but the paper argues this case is empirically unlikely.
- Quantitatively, the intertemporal-hedging component from labor income is small relative to the main mean-variance component.

4 Identification and instruments (what makes the estimates credible)

This is not an IV paper. So "identification" here means: what in the model lets the authors *separate* the different welfare channels and give their quantitative exercise content.

4.1 What fails in the naive approach

- A naive calculation looks only at the static portfolio loss from replacing the familiar portfolio by a diversified one, holding the risky share and consumption fixed.
- In that exercise, the welfare loss looks modest: around 0.77% per year in the baseline partial-equilibrium mean-variance calculation.
- The paper argues this is incomplete because it suppresses the two margins through which households actually react: the risky share π_h and the consumption rate c_h.

4.2 What identifies the new channels

- **Risky-share channel.** Equation for π_h shows that familiarity affects the total risky share through μ_{fh}.
- **Consumption-savings channel.** Equation for c_h shows that changes in the household's effective mean-variance trade-off alter optimal consumption today versus saving for tomorrow.
- **Growth channel.** In general equilibrium, aggregate saving maps into capital accumulation and hence into g.

4.3 Why the aggregate restrictions matter

- The assumption that portfolio tilts cancel out across households rules out the trivial story that some firms receive too much capital in aggregate.
- The symmetry condition on average familiarity rules out aggregate risky-share heterogeneity across households.
- Therefore, the general-equilibrium effect is a pure consequence of distorted intertemporal choices, not aggregate misallocation toward specific firms.

4.4 What disciplines the quantitative exercise

- The size of underdiversification comes from observed household portfolio volatilities.
- The risky-share calibration comes from the observed fraction of financial wealth invested in risky assets.
- The preference parameters are set to values standard in macro-finance and then varied in sensitivity analysis.
- The point of the exercise is thus comparative: how much bigger is the welfare loss when each additional channel is turned on?

4.5 Relation to the literature

- Relative to Calvet, Campbell, and Sodini, the paper says static mean-variance welfare calculations materially understate losses.
- Relative to household-finance work on mistakes, it shows that micro mistakes can survive aggregation and affect growth.
- Relative to macro-finance representative-agent models, it gives a clean channel through which household portfolio mistakes enter the real economy.

5 Tables and figures

5.1 Table 1: Individual household welfare gain

Read:

- This is the paper's core **partial-equilibrium** result.
- Start with the familiar portfolio volatility dropping from 19.5% to 14.7%.
- If only diversification is repaired while the risky share stays biased, the mean-variance gain is small:

  0.0077 in levels, 14.00% in wealth equivalent, 0.77% per year.

- If both diversification and the risky share are repaired, the mean-variance gain rises to

  0.0135, 24.63%, 1.35% per year.

- Once lifetime utility is used and intertemporal consumption is allowed to adjust, the gain becomes much larger. The fully unbiased case gives

  0.0568, 76.78%, 4.20% per year.

- So the paper's message is that the static diversification loss is only the first layer. The dynamic household response is much more important.

Table 1—Individual Household Welfare Gain

	In levels	In percent	In percent × $(1/T)$
Panel A. Mean-variance welfare gain			
$\mathbf{x}_h$ unbiased, π_h biased	0.0077	14.00	0.77
$\mathbf{x}_h$ and π_h unbiased	0.0135	24.63	1.35
Panel B. Lifetime utility welfare gain			
$\mathbf{x}_h$ unbiased, π_h and c_h biased	0.0273	36.90	2.02
$\mathbf{x}_h$ and π_h unbiased, c_h biased	0.0560	75.80	4.15
$\mathbf{x}_h$, π_h, and c_h, unbiased	0.0568	76.78	4.20

Notes: In this table, we report the increase in welfare of an individual household in partial equilibrium from reducing the effect of familiarity biases on: portfolio-diversification ($\mathbf{x}_h$), asset-allocation (π_h), and the consumption rate (c_h). Panel A measures welfare using mean-variance utility while panel B measures welfare using intertemporal lifetime utility. The parameter values used are: $\sigma_{1/N} = 14.7$ percent p.a.; $\sigma_{\mathbf{x}_h} = 19.5$ percent p.a.; $\alpha = 10.4$ percent p.a.; $i = 3.7$ percent p.a.; $\gamma_h = 3.3245$; $\delta_h = 0.03$ p.a.; $\psi_h = 1.20$.

2.png

5.2 Table 2: Social welfare gains from household portfolio diversification

Read:

- This is the paper's **general-equilibrium** counterpart to Table 1.
- **Full removal of familiarity bias (Panel A):**
  - mean-variance social gain:
  $$0.0145, \qquad 20.51\%, \qquad 1.45\% \text{ per year;}$$
  - with intertemporal consumption but exogenous growth:
  $$0.1145, \qquad 79.15\%, \qquad 5.58\% \text{ per year;}$$
  - with endogenous growth:
  $$0.1491, \qquad 103.05\%, \qquad 7.26\% \text{ per year.}$$
- **Partial removal of familiarity bias (Panel B):**
  - mean-variance gain:
  $$0.0070, \qquad 9.92\%, \qquad 0.70\% \text{ per year;}$$
  - exogenous-growth lifetime gain:
  $$0.0455, \qquad 31.46\%, \qquad 2.22\% \text{ per year;}$$
  - endogenous-growth social gain:
  $$0.0566, \qquad 39.11\%, \qquad 2.76\% \text{ per year.}$$
- The big point is that the macro growth channel adds meaningfully on top of the already large micro intertemporal-consumption channel.

5.3 Figure 1: Social welfare with changes in δ and ψ

Read:

- Panel A varies impatience δ.
- The mean-variance gain is flat because it does not depend on intertemporal preferences.
- As households become more patient (lower δ), the dynamic private gain and the social gain rise sharply.
- This shows that the amplification is fundamentally about intertemporal smoothing.
- Panel B varies the elasticity of intertemporal substitution ψ.
- At $\psi = 1$, the private dynamic gain and the social gain intersect because the macro growth effect is exactly zero.
- For $\psi > 1$, the social-gain curve lies above the private-gain curve because lower portfolio risk increases saving and growth.
- For $\psi < 1$, the growth effect works the other way, but welfare still improves.

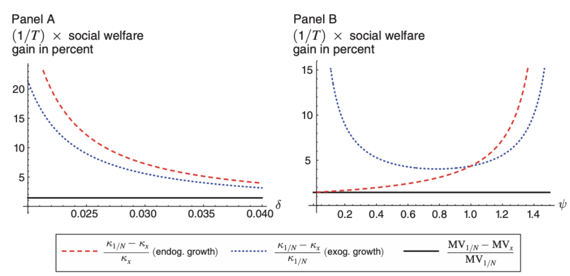

FIGURE 1. SOCIAL WELFARE WITH CHANGES IN δ AND ψ

Notes: In panel A, we plot welfare as the impatience parameter, δ, varies and in panel B, we plot welfare as the elasticity of intertemporal substitution, ψ, varies. In both panels, we report the social welfare gain per unit capital stock in percentage terms based on an annualized return over $T = 14.18$ years. Each panel has three curves: the solid (black) line shows the welfare gains from diversification for a mean-variance household; the dotted (blue) line shows the gains from portfolio diversification for a household with intertemporal consumption but with growth being exogenous; and the dashed (red) line shows the welfare gain from portfolio diversification when growth is endogenous.

TABLE 3—WELFARE GAINS WHEN δ_h, ψ_h, OR γ_h ARE HETEROGENEOUS

	In levels	In percent	In percent × $(1/T)$
Panel A. Base case without heterogeneity			
Mean-variance welfare gain			
x less biased	0.0070	9.92	0.70
Lifetime utility welfare gain			
x, c less biased but growth exogenous	0.0455	31.46	2.22
x, c less biased and growth endogenous	0.0566	39.11	2.76
Panel B. Heterogeneity in subjective discount rates			
Mean-variance welfare gain			
x unbiased	0.0070	9.92	0.70
Lifetime utility welfare gain			
x, c unbiased but growth exogenous	0.0965	37.29	2.63
x, c unbiased and growth endogenous	0.1231	46.98	3.31
Panel C. Heterogeneity in elasticity of intertemporal substitution			
Mean-variance welfare gain			
x unbiased	0.0070	9.92	0.70
Lifetime utility welfare gain			
x, c unbiased but growth exogenous	0.0628	36.13	2.55
x, c unbiased and growth endogenous	0.0919	49.72	3.51
Panel D. Heterogeneity in relative risk aversion			
Mean-variance welfare gain			
x unbiased	0.0090	12.80	0.90
Lifetime utility welfare gain			
x, c unbiased but growth exogenous	0.0618	42.71	3.01
x, c unbiased and growth endogenous	0.0777	53.71	3.79

Notes: In this table, we report the potential gains to social welfare from reducing households' familiarity biases partially. Our results are reported in four panels. Panel A gives the results for the case where there is no heterogeneity. Panels B, C, and D are for the cases where there is heterogeneity in subjective discount rates, elasticities of intertemporal substitution, and relative risk aversion levels, respectively. Heterogeneity of 1,001 households is modeled as a uniform distribution centered on the parameter value for the base case of that parameter in panel A, in which $\delta = 0.03$ p.a., $\psi = 1.20$, and $\gamma = 1.762$. In panel B, δ_h ranges from 0.02 to 0.04. In panel C, ψ_h ranges from 0.90 to 1.50. In panel D, γ_h ranges from 1.262 to 2.262. The parameter values we have assumed are: $\sigma_{1/N} = 14.7$ percent p.a.; $\sigma_{\mathbf{x}} = 19.5$ percent p.a.; and $\alpha = 10.4$ percent p.a.

5.4 Table 3: Welfare gains with preference heterogeneity

Read:

- This table asks whether preference heterogeneity makes the social cost of underdiversification larger.
- In the base case without heterogeneity, the partially corrected endogenous-growth social gain is

$$39.11\% \quad (2.76\% \text{ per year}).$$

- With heterogeneity in subjective discount rates, the same number rises to

$$46.98\% \quad (3.31\% \text{ per year}).$$

- With heterogeneity in EIS, it rises further to

$$49.72\% \quad (3.51\% \text{ per year}).$$

- With heterogeneity in risk aversion, it becomes

$$53.71\% \quad (3.79\% \text{ per year}).$$

- The paper's interpretation is straightforward: heterogeneous households have an even stronger desire for risk sharing and consumption smoothing, so underdiversification is even more costly in social terms.

5.5 Discussion of modeling choices

Read:

- The paper explicitly discusses labor income, employer stock, pension restrictions, and other background risks.
- The main argument is that because the paper studies *differences in differences* in welfare when extra channels are added, many changes to the baseline model wash out.
- So the exact level of welfare can move, but the central amplification result is meant to be robust.

6 Short summary

- **Conclusion.** Household underdiversification looks modest only in a static portfolio problem. In a dynamic general-equilibrium setting, it becomes much more costly.
- **Main theoretical insight.** Familiarity bias distorts three objects: the risky portfolio composition, the risky share, and the intertemporal consumption path.
- **Main macro insight.** Even when biased portfolio tilts cancel across households, the induced saving distortions need not cancel. Aggregate growth then changes.
- **Quantitative lesson.** A loss that looks like about 1% per year in mean-variance terms can become 4%–7% per year once intertemporal and general-equilibrium channels are included.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the welfare consequences of household portfolio mistakes are much larger than static mean-variance calculations suggest.

- The familiar result that diversification mistakes are "small" is true only if one freezes the household's risky share and consumption path and ignores general equilibrium.
- Once those margins are allowed to react, household finance matters a great deal.

7.2 Economic intuition

- A household that concentrates in familiar stocks bears too much idiosyncratic risk.
- To cope with this self-created risk, it invests too little in risky productive assets overall.
- Because the household is also intertemporal, this lower effective return changes its consumption-saving choice.
- In the aggregate, that changes capital accumulation and growth.
- So a portfolio mistake becomes a macro distortion.

7.3 Takeaway

This paper is best read as a warning against evaluating household financial mistakes with a purely static lens. The paper's answer to the title question is yes: household finance matters, because small individual portfolio errors can generate large private and social welfare costs once they distort saving and growth. For policy, the paper suggests that better default portfolios, financial education, and financial innovation that reduce familiarity bias may deliver benefits far beyond the household making the initial mistake.